

Expressions, Formulae and Brackets — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. Where a question is worth more than one mark, award method marks even when the final answer is wrong — and take them away for a correct answer with no working only if the question said "show your working". The notes under each answer flag the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS • NON-CALCULATOR

- 1** $5x + 3 - 2x + 7 \rightarrow 3x + 10$ and 16 [6]
 $8x - 3 - 5x + 9 \rightarrow 3x + 6$ and 12
 $4x + 2x - 7 + 1 \rightarrow 6x - 6$ and 6
One mark per box. Award the value box on a correct follow-through from a wrong simplification — the student is being tested on two separate skills and should not lose both for one error.
- 2** (a) $4p + 6q$ [3]
 (b) $24mn$
 (c) $15c^2$
(a) stops there; $10pq$ scores nothing. (b) and (c) make the contrast that matters — you may always multiply unlike terms, you may only add like ones. In (c), $c \times c$ is c^2 , not $2c$.
- 3** (a) $3n + 5$ [3]
 (b) $n - 7$
 (c) $2(n + 4)$ or $2n + 8$
(b) must be $n - 7$, not $7 - n$ — "decreased by" keeps the number first. In (c) the word "sum" forces the bracket; $2n + 4$ is a different expression and scores nothing.
- 4** (a) $10 + (-3) = 7$ [4]
 (b) $-9 - 5 = -14$
 (c) $5 \times (-3) = -15$
 (d) $5 - (-3) = 8$
(d) is the one to watch — subtracting a negative adds, so the answer is larger than a. A student answering 2 has treated $-(-3)$ as -3 .
- 5** (a) $3x + 12$ [4]
 (b) $10x - 15$
 (c) $12x + 8$
 (d) $10 - 2x$
Watch for the second term being left unmultiplied — $3x + 4$ in (a) or $10x - 3$ in (b). In (d) the terms are the other way round, which catches students who multiply by position rather than by meaning.
- 6** (a) $3(2x + 3)$ [4]
 (b) $5(2x - 3)$
 (c) $4(2x + 5)$
 (d) $7(2x + 3)$
Every one must be the highest common factor. $2(4x + 10)$ in (c) is not completely factorised and scores nothing. Get the student to expand each answer back — it takes seconds and settles every dispute.

- 7 (a) $40 + 12d$ rupees [1] [3]
 (b) $40 + 12(8) = 136$ rupees [1]
 (c) $40 + 12(15) = 220$ rupees [1]
In (a), $52d$ is the error to look for — it adds the fixed charge to the per-kilometre charge as though the 40 were also charged every kilometre.
- 8 (a) $2(w + w + 6) = 2(2w + 6) = 4w + 12$ cm [1] [3]
 (b) $4(5) + 12 = 32$ cm [1]
 (c) $w(w + 6) = w^2 + 6w$ cm² [1]
Check (b) without algebra: at $w = 5$ the rectangle is 5 by 11, so the perimeter is 32. In (c), $w \times w$ must become w^2 ; $2w$ is the error.

Paper B — Medium

30 MARKS · NON-CALCULATOR

- 1 $2x + 5 \rightarrow 11$ and 1 [6]
 $x^2 \rightarrow 9$ and 4
 $3x - 4 \rightarrow 5$ and -10
One mark per box. The x^2 row at $x = -2$ is the whole point: $(-2)^2 = 4$, and a student answering -4 has squared the 2 and kept the sign outside.
- 2 (a) $4y - 5$ [2]
 (b) $4ab$
In (a) the constant terms are $+3$ and -8 , giving -5 . In (b), ab is a single letter part, so those two terms really are alike — some students refuse to join them because there are two letters.
- 3 (a) $3(n + 5)$ or $3n + 15$ [2]
 (b) $2n^2$
(b) must be $2n^2$, not $(2n)^2 = 4n^2$. The order in the sentence is squared first, doubled second, and the notation follows it exactly.
- 4 (a) $2x + 6 + 3x + 3$ [1] = $5x + 9$ [1] [4]
 (b) $8x - 4 + 3x + 15$ [1] = $11x + 11$ [1]
The first mark is for expanding both brackets correctly, the second for collecting. A student who expands right and collects wrong still earns half — make sure both lines are written out.
- 5 (a) $4(3x + 2)$ [4]
 (b) $5y(y + 2)$
 (c) $3(3x - 4)$
 (d) $2a(2a + 3)$
(b) and (d) need the letter taken out as well as the number. $5(y^2 + 2y)$ is the standard half-answer and is not completely factorised.
- 6 (a) $x^2 + 7x$ [4]
 (b) $6x^2 - 2x$
 (c) $x^2 + 5x + 2x + 10 = x^2 + 7x + 10$
 (d) $6x + 3$
(c) has four products, not two — every term in the first bracket meets every term in the second. Note that (a) and (c) both give $7x$, which is a coincidence worth pointing out so it is not mistaken for a rule.
- 7 (a) $x + (x + 3) + (2x - 1)$ [1] = $4x + 2$ cm [1] [4]
 (b) $4(5) + 2 = 22$ cm [1]
 Check: sides are 5, 8 and 9, and $5 + 8 + 9 = 22$ □ [1]
The check is a mark in its own right and the question asked for it. It is also the habit worth building — every algebra-in-shapes answer can be tested against real numbers in about fifteen seconds.

- 8 (a) $4p + 3n$ [1] [4]
 (b) $5(4p + 3n) = 20p + 15n$ [1]
 (c) $20(12) + 15(45) = 240 + 675$ [1] = 915 rupees [1]
In (b) the whole bracket is multiplied by 5, not just the first term — $20p + 3n$ is the error. Accept either form in (b), but (c) is far easier from the expanded one.

Paper C — Hard

30 MARKS · NON-CALCULATOR

- 1 (a) $10x + 15 - 2x - 8$ [1] = $8x + 7$ [1] [4]
 (b) $12x - 8 - 6x + 15$ [1] = $6x + 7$ [1]
(b) is the sign test: $-3 \times -5 = +15$. A student answering $6x - 23$ has written -15 , and that single sign is worth checking on every question of this shape.
- 2 (a) $6x - 15 - x + 4$ [1] = $5x - 11$ [1] [4]
 (b) $8x + 2 - 5x + 15$ [1] = $3x + 17$ [1]
In (a) the bare minus in front of $(x - 4)$ means -1 , so both terms flip: $-x$ and $+4$. Students who ignore an invisible 1 tend to write $-x - 4$.
- 3 (a) $(-3)^2 = 9$ [4]
 (b) $2 \times 9 = 18$
 (c) $(-3 + 4)^2 = 1^2 = 1$
 (d) $9 + 16 = 25$
(c) and (d) are the same two letters and differ only by a bracket, giving 1 against 25. Any student who thinks $(a + b)^2$ and $a^2 + b^2$ are the same thing has just been shown otherwise in one line.
- 4 (a) $6(3x + 4)$ [4]
 (b) $5y(3y - 4)$
 (c) $6a(2a + 3)$
 (d) It cannot be factorised. 8 and 3 share no common factor, and x and y are different letters, so there is nothing common to every term.
(d) is a real answer, not a trick. Students who cannot bear to leave a question “undone” will invent something like $x(8 + 3y/x)$ — the reason matters more than the answer here.
- 5 (a) HCF of 20 and 45 is 5 [1], so $5(4x + 9)$ [1] [4]
 (b) The area is $5 \times (4x + 9)$, and one side is 5, so the other is $(4x + 9)$ m [1]
 (c) Perimeter = $2(5 + 4x + 9) = 2(4x + 14) = 8x + 28$ m [1]
Factorising is doing real work here — it is the only way to find the missing side. Worth saying out loud, because students often see factorising as a pointless rearrangement.
- 6 (a) $n + (n + 1) = 2n + 1$ [1] [5]
 (b) $2n$ is a multiple of 2 and so is always even [1]; one more than an even number is always odd, so the sum is always odd [1]
 (c) $n + (n + 1) + (n + 2) = 3n + 3$, which factorises to $3(n + 1)$ [1]; that is 3 times a whole number, so it is always a multiple of 3 [1]
The explanation carries the marks, not the algebra. “I tried 4 and 5 and got 9, which is odd” is a check, not a proof, and earns nothing in (b) — the argument must be about $2n$ being even for every n .
- 7 (a) $150 + 25(6) = 300$ rupees [1] [5]
 (b) $425 - 150 = 275$ [1]; $275 \div 25 = 11$ hours [1]
 (c) $210 - 150 = 60$ [1]; $60 \div 25 = 2.4$, which is not a whole number of hours, so the cost 210 is impossible [1]
(c) is the question that separates understanding from procedure. The point is that every possible cost is 150 plus a multiple of 25, and 210 is not — accept that argument as an alternative to the division.

- 1** (a) $12x - 18 - 4x + 20$ [1] = $8x + 2$ [1] [4]
 (b) $12x + 6 - 6x + 14$ [1] = $6x + 20$ [1]

Both hinge on a negative times a negative. In (b), $-2 \times -7 = +14$; a student writing -14 gets $6x - 8$ and will make the same error again next week unless it is named.

- 2** (a) $3 \times 4 = 12$ [4]
 (b) $(-6)^2 = 36$
 (c) $5 - (-2) = 7$
 (d) $-8 + 5 = -3$

(a) against (b) is the bracket trap in its cleanest form: square then multiply gives 12, multiply then square gives 36. (d) tests that an odd power of a negative stays negative.

- 3** (a) $12(2x + 3)$ [4]
 (b) $7y(2y - 3)$
 (c) $15(2a + 3b)$
 (d) $3(2x^2 + 3x + 4)$

(d) has three terms, and the common factor must divide all three — 3 does, but no letter does, because the last term has none. Students often try to take out an x and are stopped by the 12.

- 4** (a) Perimeter is twice the sum of two adjacent sides, so that sum is $(14x + 6) \div 2 = 7x + 3$ cm [1]
 (b) $(7x + 3) - (2x + 5)$ [1] = $7x + 3 - 2x - 5 = 5x - 2$ cm [1]
 (c) At $x = 3$ the sides are 11 cm and 13 cm, so the perimeter is $2(11 + 13) = 48$ cm, and $14(3) + 6 = 48$ [1]

In (b) the whole second bracket is subtracted, so both of its terms change sign. A student writing $5x + 8$ has subtracted only the 2x. The check in (c) would have caught it, which is exactly why it is asked for.

- 5** (a) $2n + 6$ [1] [4]
 (b) $2n + (2n + 2) + (2n + 4) = 6n + 6$ [1]
 (c) $6n + 6 = 6(n + 1)$ [1]; that is 6 times a whole number, so the sum is always a multiple of 6 [1]

The final mark needs the sentence, not just the factorisation. Note the result is stronger than students expect — not merely even, but always divisible by 6.

- 6** Student R expanded the first bracket correctly but multiplied -2 by -4 and wrote -8 [1]. [5]
 Two negatives multiply to a positive, so that term should be $+8$ [1]
 Student S multiplied only the first term inside each bracket, leaving the 3 and the 4 untouched [1]. The number outside multiplies every term inside [1]
 Correct: $10x - 15 - 2x + 8 = 8x - 7$ [1]

These are the two failure modes of expanding, and it is worth showing they are independent: R multiplied everything but got one sign wrong, S got every sign right but did not multiply everything. Different fixes.

- 7** (i) The mistake: $3x$ and 2 are not like terms and cannot be added [1]. Correct: $3x + 2$ is already fully simplified — some expressions simply stop [1] [5]
 (ii) The mistake: 2 is a common factor but not the highest one — 6 and 9 inside the bracket still share a 3 [1]. Correct: $6(2x + 3)$ [1]
 The check: look inside the finished bracket. If those numbers still share a factor, the factorisation is incomplete [1]

The last mark is the transferable one. A student who can state that check will never hand in a partly factorised answer again — and it is the single most commonly repeated error in this topic.